Problem Set 1
Simple Regression Model
Econometrics

1. The following table contains the ACT (equivalent to EVAU) scores and the GPA (grade point average) for eight college students. Grade point average is based on a four-point scale and has been rounded to one digit after the decimal.

Student	GPA	ACT
1	2.8	21
2	3.4	24
3	3.0	26
4	3.5	27
5	3.6	29
6	3.0	25
7	2.7	25
8	3.7	30

(a) Estimate the relationship between GPA and ACT using OLS; that is, obtain the intercept and slope estimates in the equation

$$\widehat{GPA} = \hat{\beta}_0 + \hat{\beta}_1 ACT$$

Comment on the direction of the relationship. Does the intercept have a useful interpretation here? Explain. How much higher is the GPA predicted to be if the ACT score is increased by five points?

(b) Compute the fitted values and residuals for each observation, and verify that the residuals (approximately) sum to zero.

(c) What is the predicted value of GPA when $ACT = 20$?

(d) How much of the variation in GPA for these eight students is explained by ACT ? Explain.

Solution:

(a) Let $y_i = GPA_i$, $x_i = ACT_i$, and $n = 8$. Then:

$$\bar{x} = 25.875, \quad \bar{y} = 3.2125, \quad \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}) = 5.8125, \quad \text{and} \quad \sum_{i=1}^{n} (x_i - \bar{x})^2 = 56.875.$$

We obtain the slope as:

$$\hat{\beta}_1 = \frac{5.8125}{56.875} \approx 0.1022,$$

rounded to four decimal places. The intercept is calculated as:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \approx 3.2125 - (0.1022 \times 25.875) \approx 0.5681.$$

So, we can write the estimated regression equation as:

$$\widehat{GPA} = 0.5681 + 0.1022 \times ACT.$$

The intercept does not have a useful interpretation because ACT is not close to zero for the population of interest. If ACT is 5 points higher, $\widehat{GPA}$ increases by $.1022(5) = .511$.

(b) The fitted values and residuals - rounded to four decimal places - are given along with the observation number i and GPA in the following table:

i	GPA	$\widehat{GPA}$	$\hat{u}$
1	2.8	2.7143	.0857
2	3.4	3.0209	.3791
3	3.0	3.2253	-.2253
4	3.5	3.3275	.1725
5	3.6	3.5319	.0681
6	3.0	3.1231	-.1231
7	2.7	3.1231	-.4231
8	3.7	3.6341	.0659

You can verify that the residuals, as reported in the table, sum to -.0002 , which is pretty close to zero given the inherent rounding error.

(c) When $ACT = 20, \widehat{GPA} = .5681 + .1022(20) \approx 2.61$.

(d) The sum of squared residuals, $\sum_{i=1}^{n} \hat{u}_i^2$, is about 4347 (rounded to four decimal places), and the total sum of squares, $\sum_{i=1}^{n} (y_i - \bar{y})^2$, is about 1.0288. So the R-squared from the regression is

$$R^2 = 1 - \text{SSR/SST} \approx 1 - (.4347/1.0288) \approx .577$$

Therefore, about 57.7% of the variation in GPA is explained by ACT in this small sample of students.

2. You have collected the folloing information about variables y − x:

$\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})$	-0.5
$\sum_{i=1}^{n} (x_i - \bar{x})^2$	2
$\bar{y}$	4
$\bar{x}$	3
$\sum_{i=1}^{n} (y_i - \bar{y})^2$	0.8
$\sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2$	0.5
$\sum_{i=1}^{n} (y_i - \hat{y}_i)^2$	0.3

(a) Estimate and interpret the coeficients of the following relation:

$$y = \beta_0 + \beta_1 x + u$$

(b) Calculate and interpret the R^2

Solution:

(a) Given:

$$\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}) = -0.5, \quad \sum_{i=1}^{n} (x_i - \bar{x})^2 = 2, \quad \bar{y} = 4, \quad \bar{x} = 3.$$

Step 1: Calculate the slope $\hat{\beta}_1$:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2} = \frac{-0.5}{2} = -0.25.$$

Step 2: Calculate the intercept $\hat{\beta}_0$:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \cdot \bar{x} = 4 - (-0.25) \times 3 = 4 + 0.75 = 4.75.$$

Therefore, the estimated relationship is:

$$\hat{y} = 4.75 - 0.25x.$$

Interpretation:

- The slope $\hat{\beta}_1 = -0.25$ indicates that for each unit increase in x, y decreases by 0.25 units.
- The intercept $\hat{\beta}_0 = 4.75$ suggests that when $x = 0$, the predicted value of y is 4.75. However, the interpretation of the intercept depends on the context of the variables.

(b) Given:

$$\sum_{i=1}^{n} (y_i - \bar{y})^2 = 0.8, \quad \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2 = 0.5, \quad \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = 0.3.$$

The R^2 is calculated as:

$$R^2 = \frac{\sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2} = \frac{0.5}{0.8} = 0.625.$$

Interpretation: An R^2 value of 0.625 indicates that approximately 62.5% of the variability in y (dependent variable) is explained by x (independent variable) in the model. This suggests a moderately strong relationship between the variables.

3. You have estimated the following regression models for the relationship between the salary of a CEO (salary) and the financial return on equity (roe):

$$\begin{aligned}
\text{salary} &= 963.2 + 18.5 \times \text{ roe} \\
\log(\widehat{\text{salary}}) &= 6.71 + 0.014 \times \text{ roe} \\
\log(\widehat{\text{salary}}) &= 6.49 + 0.17 \times \log(\text{ roe }) \\
\widehat{\text{salary}} &= 586.6 + 255.3 \times \log(\text{ roe })
\end{aligned}$$

Interpret the estimated slope coefficient of each regression.

Solution:

(a) **First Regression**: $\text{salary} = 963.2 + 18.5 \times \text{roe}$

The slope coefficient 18.5 means that for every one-unit increase in roe, the salary of the CEO is expected to increase by 18.5 units (in the same currency as salary). This is a linear relationship.

(b) **Second Regression**: $\log\left(\widehat{\text{salary}}\right) = 6.71 + 0.014 \times \text{roe}$

The slope coefficient 0.014 means that for every one-unit increase in roe, the logarithm of the CEO's salary is expected to increase by 0.014. In percentage terms, a one-unit increase in roe corresponds to approximately a 1.4% increase in salary (since the percentage change is approximately 100×0.014).

(c) **Third Regression**: $\log(\widehat{\text{salary}}) = 6.49 + 0.17 \times \log(\text{roe})$

The slope coefficient 0.17 means that for a 1% increase in roe, the CEO's salary is expected to increase by 0.17%. This indicates a logarithmic-logarithmic relationship between salary and roe.

(d) **Fourth Regression**: $\widehat{\text{salary}} = 586.6 + 255.3 \times \log(\text{roe})$

The slope coefficient 255.3 means that for every 1% increase in roe, the salary of the CEO is expected to increase by approximately 255.3 units (in the same currency as salary).

4. You have estimated the following relationship between monthly net salary in Euros and the gender of a person (where "woman" is a binary variable equal to 1 if the observation is female and 0 if male):

$$\widehat{\text{salary}} = 1200 - 100 \times \text{woman}$$

(a) Interpret the OLS estimate of the intercept coefficient.

(b) Interpret the OLS estimate of the slope coefficient:

(c) What is the predicted salary for a woman?

Solution:

(a) The intercept 1200 represents the predicted salary for a male (since "woman" equals 0 for males). Therefore, the intercept indicates that the average predicted salary for a male is 1200 Euros per month.

(b) The slope coefficient -100 indicates that, on average, women earn 100 Euros less than men, holding everything else constant. If the value of "woman" is 1 (indicating female), the predicted salary decreases by 100 Euros.

(c) To find the predicted salary for a woman, substitute woman $= 1$ into the regression equation:

$$\widehat{\text{salary}} = 1200 - 100 \times 1 = 1200 - 100 = 1100 \text{ Euros.}$$

Thus, the predicted salary for a woman is 1100 Euros per month.

5. In the simple linear regression model $y = \beta_0 + \beta_1 x + u$, suppose that $\mathrm{E}(u) \neq 0$. Letting $\alpha_0 = \mathrm{E}(u)$, show that the model can always be rewritten with the same slope, but a new intercept and error, where the new error has a zero expected value.

Solution:

Given the simple linear regression model:

$$y = \beta_0 + \beta_1 x + u,$$

where $\mathrm{E}(u) \neq 0$, we need to show that the model can be rewritten with the same slope but a new intercept and error term, where the new error has a zero expected value.

Step 1: Define the Expected Value of u:

Let:

$$\alpha_0 = \mathrm{E}(u).$$

By definition, the expected value of u is α_0, meaning $\mathrm{E}(u) = \alpha_0$.

Step 2: Decompose u into Two Parts:

We can express u as:

$$u = \alpha_0 + \tilde{u},$$

where $\tilde{u}$ is a new error term with $\mathrm{E}(\tilde{u}) = 0$. Notice that $\tilde{u} = u - \alpha_0$ by construction.

Step 3: Substitute $u = \alpha_0 + \tilde{u}$ into the Original Model:

Substituting into the original equation gives:

$$y = \beta_0 + \beta_1 x + u = \beta_0 + \beta_1 x + \alpha_0 + \tilde{u}.$$

Step 4: Rearrange the Model:

We can now rewrite the model as:

$$y = (\beta_0 + \alpha_0) + \beta_1 x + \tilde{u}.$$

Let the new intercept be:

$$\beta_0' = \beta_0 + \alpha_0.$$

So the model becomes:

$$y = \beta_0' + \beta_1 x + \tilde{u},$$

where $\tilde{u}$ is the new error term.

Step 5: Verify the Expected Value of the New Error Term:

By construction, $\tilde{u} = u - \alpha_0$, and:

$$\mathrm{E}(\tilde{u}) = \mathrm{E}(u - \alpha_0) = \mathrm{E}(u) - \alpha_0 = \alpha_0 - \alpha_0 = 0.$$

Conclusion:

The model has the same slope β_1, but a new intercept $\beta_0' = \beta_0 + \alpha_0$, and the new error term $\tilde{u}$ has an expected value of zero $\mathrm{E}(\tilde{u}) = 0$.

6. Suppose you are interested in estimating the effect of hours spent in a final exam preparation (in hours) on the final grade. The population is all students enrolled in that course.

   (a) Suppose you are given a grant to run a controlled experiment. Explain how you would structure the experiment in order to estimate the causal effect of hours on sat.

   (b) Consider the more realistic case where students choose how much time to spend in a preparation course, and you can only randomly sample grade and hours from the population. Write the population model as

   $$\text{grade} = \beta_0 + \beta_1 \text{hours} + u$$

   where, as usual in a model with an intercept, we can assume $\mathrm{E}(u) = 0$. List at least two factors contained in u. Are these likely to have positive or negative correlation with hours?

   (c) What should be the sign of β_1 if the preparation course is effective?

   (d) What is the interpretation of β_0 ?

   Solution:

   (a) We would want to randomly assign the number of hours in the preparation course so that hours is independent of other factors that affect performance in terms of grade. Then, we would collect information on grade for each student in the experiment, yielding a data set $\{($ grade $_i$, hours $_i) : i = 1, \ldots, n\}$, where n is the number of students we can afford to have in the study. We should try to get as much variation in hours $_i$ as is feasible.

   (b) Here are two factors: innate ability and general health on the day of the exam. If we think students with higher native intelligence think they do not need to prepare for the exam, then ability and hours will be negatively correlated. Ruling out chronic health problems, health on the day of the exam should be roughly uncorrelated with hours spent in a preparation course.

   (c) If preparation courses are effective, β_1 should be positive: other factors equal, an increase in hours should increase grade.

   (d) The intercept, β_0, has a useful interpretation in this example: because $\mathrm{E}(u) = 0, \beta_0$ is the average grade for students in the population with hours $= 0$.

7. According to the capital asset pricing model (CAPM), the expected excess return on an asset is proportional to the expected excess return on a portfolio of all available assets (the "market portfolio"). That is, the CAPM says that

   $$R - R_f = \beta \left(R_m - R_f\right)$$

   where R_m is the expected return on the market portfolio and β is the coefficient in the population regression of $R - R_f$ on $R_m - R_f$. In practice, the risk-free return is often taken to be the rate of interest on short-term U.S. government debt. According to the CAPM. a stock with a $\beta < 1$ has less risk than the market portfolio and therefore has a lower expected excess return than the market portfolio. In contrast, a stock with a $\beta > 1$ is riskier than the market portfolio and thus commands a higher expected excess return.

   (a) Suppose that the value of β is greater than 1 for a particular stock. Show that the variance of $(R - R_f)$ for this stock is greater than the variance of $(R_m - R_f)$.

(b) Suppose that the value of β is less than 1 for a particular stock. Is it possible that variance of $(R - R_f)$ for this stock is greater than the variance of ($R_m - R_t$) ? (Hint: Don't forget the regression error.)

(c) In a given year, the rate of return on 3-month Treasury bills is 3.5% and the rate of return on a large diversified portfolio of stocks (the S&P 500) is 7.3%. For each company listed in the following table use the estimated value of β to estimate the stock's expected rate of return.

Company	Estimated β
Kellogg (breakfast cereal)	-0.03
Wall-Mart (discount retailer)	0.65
Waste Management (Waste disposal)	0.70
Sprint Nextel (telecommunications)	0.78
Barnes and Noble (book retailer)	1.02
Microsoft (software)	1.27
Best Buy (electronic equipment retailer)	2.15
Amazon (online retailer)	2.65

Solution:

(a) We need to show that if $\beta > 1$, then the variance of $(R - R_f)$ is greater than the variance of $(R_m - R_f)$ for a particular stock. The variance of $(R - R_f)$ is given by:

$$\text{Var}(R - R_f) = \text{Var}(\beta(R_m - R_f) + u).$$

Using the properties of variance, we can express the variance as:

$$\text{Var}(R - R_f) = \beta^2\text{Var}(R_m - R_f) + \text{Var}(u).$$

Notice that $\beta^2\text{Var}(R_m - R_f)$ is the part of the variance that is directly related to the market portfolio. Since $\beta > 1$ implies $\beta^2 > 1$, we have:

$$\beta^2\text{Var}(R_m - R_f) > \text{Var}(R_m - R_f).$$

Thus:

$$\text{Var}(R - R_f) = \beta^2\text{Var}(R_m - R_f) + \text{Var}(u) > \text{Var}(R_m - R_f).$$

Conclusion: When $\beta > 1$, the variance of $(R - R_f)$ is greater than the variance of $(R_m - R_f)$, which confirms that a stock with a higher β is indeed riskier than the market portfolio.

(b) We need to determine whether it is possible for the variance of $(R - R_f)$ to be greater than the variance of $(R_m - R_f)$ when $\beta < 1$ for a particular stock. The variance of $(R - R_f)$ is given by:

$$\text{Var}(R - R_f) = \text{Var}(\beta(R_m - R_f) + u).$$

Using the properties of variance, we can express the variance as:

$$\text{Var}(R - R_f) = \beta^2\text{Var}(R_m - R_f) + \text{Var}(u).$$

Consider the Case Where $\beta < 1$. If $\beta < 1$, then $\beta^2 < 1$, which implies:

$$\beta^2\text{Var}(R_m - R_f) < \text{Var}(R_m - R_f).$$

Notice that the total variance of $(R - R_f)$ includes the error variance $\text{Var}(u)$. Therefore:

$$\text{Var}(R - R_f) = \beta^2 \text{Var}(R_m - R_f) + \text{Var}(u).$$

Even though $\beta^2 \text{Var}(R_m - R_f)$ is less than $\text{Var}(R_m - R_f)$, the error variance $\text{Var}(u)$ could be large enough to make the overall variance $\text{Var}(R - R_f)$ greater than $\text{Var}(R_m - R_f)$.

(c) The expected returns for each company are calculated as follows:

$$\begin{aligned}
\text{Kellogg (breakfast cereal)}: &\quad 3.5\% + (-0.03) \times (7.3\% - 3.5\%) = 3.386\%, \\
\text{Wall-Mart (discount retailer)}: &\quad 3.5\% + 0.65 \times (7.3\% - 3.5\%) = 5.97\%, \\
\text{Waste Management (Waste disposal)}: &\quad 3.5\% + 0.70 \times (7.3\% - 3.5\%) = 6.16\%, \\
\text{Sprint Nextel (telecommunications)}: &\quad 3.5\% + 0.78 \times (7.3\% - 3.5\%) = 6.464\%, \\
\text{Barnes and Noble (book retailer)}: &\quad 3.5\% + 1.02 \times (7.3\% - 3.5\%) = 7.376\%, \\
\text{Microsoft (software)}: &\quad 3.5\% + 1.27 \times (7.3\% - 3.5\%) = 8.326\%, \\
\text{Best Buy (electronic equipment retailer)}: &\quad 3.5\% + 2.15 \times (7.3\% - 3.5\%) = 11.67\%, \\
\text{Amazon (online retailer)}: &\quad 3.5\% + 2.65 \times (7.3\% - 3.5\%) = 13.57\%.
\end{aligned}$$

Computer Exercises

1. The data set in CEOSAL2 contains information on chief executive officers for U.S. corporations. The variable salary is annual compensation, in thousands of dollars, and ceoten is prior number of years as company CEO.

   (a) Find the average salary and the average tenure in the sample.

   (b) How many CEOs are in their first year as CEO (that is, ceoten = 0)? What is the longest tenure as a CEO?

   (c) Estimate the simple regression model

   $$\log(\text{ salary }) = \beta_0 + \beta_1 \text{ ceoten } + u$$

   and report your results in the usual form. What is the (approximate) predicted percentage increase in salary given one more year as a CEO?

Solution:

```
# install.packages("wooldridge")

data(ceosal2, package='wooldridge')

# Calculate the average salary
average_salary <- mean(ceosal2$salary)
average_salary

# Count the number of CEOs in their first year as CEO (ceoten = 0)
num_first_year_ceos <- sum(ceosal2$ceoten == 0)
num_first_year_ceos

# Find the longest tenure as a CEO
```

```r
longest_tenure <- max(ceosal2$ceoten)
longest_tenure

# Estimate the simple regression model
model <- lm(lsalary ~ ceoten, data = ceosal2)

# Report the regression results
summary(model)

# Extract the coefficient for ceoten (beta_1)
beta_1 <- coef(model)["ceoten"]

# Calculate the approximate percentage increase in salary for one more year as
    a CEO
percentage_increase <- beta_1 * 100

# Print the results
percentage_increase
```

2. Use the data in WAGE2 to estimate a simple regression explaining monthly salary (wage) in terms of IQ score (IQ).

   (a) Find the average salary and average IQ in the sample. What is the sample standard deviation of IQ? (IQ scores are standardized so that the average in the population is 100 with a standard deviation equal to 15.)

   (b) Estimate a simple regression model where a one-point increase in IQ changes wage by a constant dollar amount. Use this model to find the predicted increase in wage for an increase in IQ of 15 points. Does IQ explain most of the variation in wage?

   (c) Now, estimate a model where each one-point increase in IQ has the same percentage effect on wage. If IQ increases by 15 points, what is the approximate percentage increase in predicted wage?

Solution:

```r
data(wage2, package='wooldridge')

# Calculate the average salary
average_salary <- mean(wage2$wage, na.rm = TRUE)

# Calculate the average IQ
average_IQ <- mean(wage2$IQ, na.rm = TRUE)

# Calculate the sample standard deviation of IQ
sd_IQ <- sd(wage2$IQ, na.rm = TRUE)

# Print the results
average_salary
average_IQ
sd_IQ

# Estimate the simple regression model
model <- lm(wage ~ IQ, data = wage2)

# View the regression results
```

```r
summary(model)

# Extract the coefficient for IQ (the predicted increase in wage for a one-
   point increase in IQ)
coef_IQ <- coef(model)["IQ"]

# Calculate the predicted increase in wage for an increase in IQ of 15 points
predicted_increase <- coef_IQ * 15

# Print the results
coef_IQ
predicted_increase

# Check the R-squared value to see if IQ explains most of the variation in wage
r_squared <- summary(model)$r.squared
r_squared

# Estimate the log-linear regression model (log of wage vs. IQ)
model_log <- lm(log(wage) ~ IQ, data = wage2)

# View the regression results
summary(model_log)

# Extract the coefficient for IQ (percentage effect on wage for a one-point
   increase in IQ)
coef_log_IQ <- coef(model_log)["IQ"]

# Calculate the approximate percentage increase in predicted wage for a 15-
   point increase in IQ
percentage_increase <- coef_log_IQ * 15 * 100

# Print the results
coef_log_IQ
percentage_increase
```

3. To complete this exercise you need to generate data from the uniform and normal distributions.

   (a) Start by generating 500 observations on x_i-the explanatory variable-from the uniform distribution with range $[0, 10]$ distribution and multiply those observations by 10. What are the sample mean and sample standard deviation of the x_i ?

   (b) Randomly generate 500 errors, u_i, from the Normal$(0, \sigma^2 = 36)$ distribution. Is the sample average of the u_i exactly zero? Why or why not? What is the sample standard deviation of the u_i ?

   (c) Now generate the y_i as

   $$y_i = 1 + 2x_i + u_i \equiv \beta_0 + \beta_1 x_i + u_i$$

   that is, the population intercept is one and the population slope is two. Use the data to run the regression of y_i on x_i. What are your estimates of the intercept and slope? Are they equal to the population values in the above equation? Explain.

   (d) Obtain the OLS residuals, $\hat{u}_i$, and verify that equations

   $$\sum_{i=1}^{n} \hat{u}_i = 0, \quad \sum_{i=1}^{n} x_i \hat{u}_i = 0.$$

holds (subject to rounding error).

Solution:

```
# Set seed for reproducibility
set.seed(123)

# Generate 500 observations from a uniform distribution on [0, 10]
x_i <- runif(500, min = 0, max = 10) * 10

# Calculate the sample mean of x_i
sample_mean <- mean(x_i)

# Calculate the sample standard deviation of x_i
sample_sd <- sd(x_i)

# Print the results
sample_mean
sample_sd

# Set seed for reproducibility
set.seed(123)

# Step 1: Generate 500 errors from the Normal(0, 36) distribution
u_i <- rnorm(500, mean = 0, sd = sqrt(36))

# Calculate the sample mean and sample standard deviation of u_i
sample_mean_u <- mean(u_i)
sample_sd_u <- sd(u_i)

# Print the sample mean and sample standard deviation of u_i
sample_mean_u
sample_sd_u

# Explanation:
# The sample average of u_i is not exactly zero because it is a random sample,
# and although the expected value is zero, the sample mean will vary around
   zero.

# Step 2: Generate y_i = 1 + 2 * x_i + u_i
y_i <- 1 + 2 * x_i + u_i

# Step 3: Run the regression of y_i on x_i
model <- lm(y_i ~ x_i)

# View the regression summary
summary(model)

# Extract the estimated intercept and slope
estimated_intercept <- coef(model)["(Intercept)"]
estimated_slope <- coef(model)["x_i"]

# Print the estimated intercept and slope
estimated_intercept
estimated_slope

# Step 1: Obtain the OLS residuals
```

```r
residuals_u_hat <- resid(model)  # OLS residuals

# Step 2: Verify that the sum of the residuals is approximately zero
sum_residuals <- sum(residuals_u_hat)

# Step 3: Verify that the sum of x_i multiplied by the residuals is
   approximately zero
sum_x_u_hat <- sum(x_i * residuals_u_hat)

# Print the results
sum_residuals
sum_x_u_hat
```